

SHASHANK MANGALAPALLY

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PROFESSIONAL SUMMARY

Software Engineer with 3 years of experience building scalable, cloud-native applications using Python, Java, and React.js. Skilled in developing APIs, microservices, and data-driven platforms with Django, FastAPI, and Flask. Proficient in SQL and PostgreSQL optimization and experienced in deploying solutions on AWS and Azure with Docker, Kubernetes, and CI/CD pipelines. Adept at implementing machine learning utilities, real-time data pipelines, and automated workflows that enhance performance and business outcomes.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, SQL, C.

Python Frameworks & Libraries: Django, FastAPI, Flask, Pandas, NumPy, Scikit-learn, TensorFlow, Matplotlib, Seaborn, Streamlit.

Backend & API Development: Spring Boot, Node.js, Express.js, REST APIs, GraphQL, Microservices Architecture.

Frontend Development: React.js, Redux, HTML5, CSS3, Bootstrap, TypeScript.

Cloud & DevOps: AWS (EC2, Lambda, S3, RDS, DynamoDB, CloudWatch), Azure, Docker, Kubernetes, Jenkins, Terraform, GitHub Actions, CI/CD Pipelines.

Databases: PostgreSQL, MySQL, Oracle, MongoDB, DynamoDB.

Software Engineering Practices: Agile/Scrum, Git, JIRA, Confluence, Test-Driven Development (TDD), Unit Testing (PyTest, JUnit), System Design, API Design, Object-Oriented Design.

PROFESSIONAL EXPERIENCE

Student worker (Junior Developer)

May 2024 - May 2025

University of New Haven | West Haven, CT

- Led weekly sessions in Python and Data Structures, guiding 30+ graduate students through hands-on coding challenges that boosted assignment success rates by 25%.
- Helped students architect REST APIs with Django, FastAPI, and Spring Boot, resulting in 20+ projects deployed on AWS Lambda and Azure Functions with a 90% success rate.
- Designed unit-tested coursework using PyTest and JUnit, embedding TDD practices into projects and driving a 35% drop in common coding errors.
- Coached students on PostgreSQL optimization, introducing indexing and query refactoring techniques that improved database performance by 30% in final submissions.
- Built cloud-based labs with Docker and AWS EC2, where 85% of participants successfully containerized and deployed applications in production-like environments.
- Mentored six capstone teams in React.js and Node.js development, enabling secure JWT-authenticated APIs and achieving a 100% deployment readiness rate.

Software Engineer

Aug 2021 - Jul 2023

Augur Talent Care | Hyderabad, India

- Developed scalable backend services and APIs using Python (FastAPI, Django), reducing average API response times by 35% and improving application responsiveness.
- Built real-time data pipelines with Python, Pandas, and SQL, processing 1M+ records daily and cutting data transformation latency by 40%, which enhanced analytics dashboards.
- Automated batch workflows with AWS Lambda and S3, reducing manual intervention by 50% and ensuring reliable execution in a cloudnative environment.
- Implemented anomaly detection modules using Scikit-learn and NumPy, increasing prediction accuracy by 18% and strengthening datadriven decision-making.
- Deployed Python microservices with Docker, Kubernetes, and Jenkins, accelerating release cycles by 45% while improving rollback efficiency.
- Optimized relational databases with PostgreSQL and MySQL query tuning, achieving 30% faster execution times and delivering quicker reporting insights.
- Elevated software reliability by enforcing unit and integration testing (PyTest, unittest), reaching 85%+ code coverage and minimizing production defects.

PROJECTS

WhatsApp Chat Analytics Platform

- Developed a Python preprocessing pipeline using regex, Unicode normalization, and language detection to extract structured chat data, achieving 95% parsing accuracy across 100K+ messages and enabling reliable NLP feature extraction.
- Delivered an interactive Streamlit dashboard with Pandas and CountVectorizer, containerized and deployed via Heroku and GitHub Actions CI/CD, cutting analysis time by 40% and supporting concurrent multi-user sessions.

Movie Recommender System

- Built content-based Movie Recommendation System in Python using TMDB 5000+ dataset, applying feature engineering on genres, keywords, cast/crew, and overview metadata with NLP preprocessing (CountVectorizer, Porter Stemming) to generate 5000-dimensional feature vectors.
- Developed cosine similarity algorithm achieving 85%+ recommendation accuracy and deployed interactive Streamlit app with TMDB API for dynamic poster fetching, responsive UI, and scalable user experience.

SkillLens - Activity Tracking & Visualization Platform

- Architected RESTful APIs in Node.js/Express with PostgreSQL, JWT authentication, and role-based access, enabling secure analytics queries and scalable activity ingestion across the platform.
- Built a React + TypeScript frontend with Redux and Chart.js, containerized services using Docker and Kubernetes, and set up AWS-based CI/CD pipelines to deliver interactive dashboards with real-time insights and ensure zero-downtime deployments.

EDUCATION

Masters of Science in Information Science

Aug 2023 - May 2025

University of New Haven | West Haven, CT

CERTIFICATIONS

- Machine Learning Specialization - **Stanford University Coursera**
- Python for Everybody Specialization - **Coursera**
- Data Structures and Algorithms in Python - **LinkedIn Learning**
- Software Design and Architecture Specialization - **Coursera**